

GxP eSignature & Reauthentication

Integration Details & Test Guide

SAP S/4HANA ME21N | BTP eSignature SaaS | Reauthentication Framework

Date: 2026-06-26 | Version: 1.0 | Status: Pre-2702 Planning

Availability Status — What Is Ready vs What Is Planned

Component	Status	Details
eSignature BTP SaaS Service	■ Available on BTP	Service exists and is ready to integrate — does NOT need to be built
Reauthentication Framework (S4)	■ Available in S4	Framework built, APIs exposed, S-table config available in SPRO
Change Document for PO	■ Done — verified 2608	CDHDR/CDPOS working for all standard PO fields including ATO extension fields
eSignature integration in ME21N	■ NOT YET BUILT	Trigger point in ME21N save flow — planned for 2702
Communication Arrangement S4↔BTP	■ NOT YET configured	OAuth / service key between S4 and BTP eSignature tenant — needed for 2702
Reauthentication in ME21N save	■ NOT YET integrated	Relevancy Check callback class + S-table registration — planned for 2702
BTP Reason Codes for PO	■ NOT YET configured	Reason codes for PURCHASE_ORDER object in BTP — part of 84P provisioning

■ Testable TODAY: Change Document for PO (verified 2608), CHECK_ESIG_ACTIVE API (returns FALSE — expected), Reauthentication S-table configuration in SPRO (framework is available).

■ Requires 2702 delivery before full testing: reauthentication dialog in ME21N, BTP eSignature SaaS connection, signature capture linked to Change Document, end-to-end GxP audit trail. Contact: Swarnava Chatterjee (eSignature) + Yadesh Gupta (Reauthentication Framework) for test system access.

Section 1 — eSignature Integration Details

1.1 Where eSignature Is Triggered in ME21N

The eSignature trigger point sits inside the ME21N save flow — AFTER Change Document fires but BEFORE the Logical Unit of Work (LUW) is committed to the database. If the user cancels the reauthentication dialog, the ENTIRE save is rolled back — nothing is committed.

ME21N SAVE FLOW — eSignature Integration Point

Step 1 User fills PO fields and clicks Save

Step 2 EKKO / EKPO written (standard PO processing)

Step 3 /ATO/S4_PO_HEADER_EXT written (ATO extension fields)

Step 4 Change Document fires → CDHDR + CDPOS created ← done 2608 ✓

Records: field, old value, new value, user, timestamp

Step 5 [NEW – 2702] CHECK_ESIG_ACTIVE

→ If FALSE: skip all reauthentication, commit normally

→ If TRUE: proceed to relevancy check

Step 6 [NEW – 2702] Relevancy Check callback: ZCL_PO_REAUTH_CHECK

→ Reads S-table VC_SRA_S0T for regulated fields

→ ABAP_FALSE: no regulated field changed → skip, commit

→ ABAP_TRUE: regulated field changed → LUW PAUSED

Step 7 [NEW – 2702] REQUEST_REAUTH (HCP GUI variant for ME21N)

→ Dialog shown: Password re-entry + Reason Code dropdown

→ User CANCELS → ROLLBACK – PO NOT saved

→ User CONFIRMS → BTP eSignature SaaS validates password

Step 8 [NEW – 2702] VALIDATE_RESULT

→ Confirms signature captured in BTP

→ Signature ID linked to CHANGENR in CDHDR

Step 9 COMMIT WORK – all writes committed atomically

Step 10 Outbound event published → ATO SaaS notified via Event Mesh

1.2 The 4 APIs Applications Must Call

API	When to Call	Technology	Status
CHECK_ESIG_ACTIVE	Before any signature flow — check if GxP is active	Any	Available
REQUEST_REAUTH (OData/GUI)	Trigger reauthentication dialog in ME21N	HCP GUI / Web GUI	2702
REQUEST_REAUTH (library)	Trigger reauthentication in all other app types	All technologies	2702
VALIDATE_RESULT	Confirm signature was successfully captured	Any technology	2702

1.3 Communication Arrangement Setup (One-Time)

Step	Activity	Details
1	Create Communication Arrangement in S4	System Alias pointing to BTP eSignature SaaS tenant
2	Configure OAuth / Service Key	Exchange credentials between S4 and BTP eSignature service
3	Test connectivity	Verify S4 can reach BTP eSignature endpoint before enabling in production
4	Pattern reference	Follows same pattern as SAP_COM_0267 for other BTP integrations

Section 2 — Reauthentication Framework Integration Details

2.1 Relevancy Check Callback — What LOB Team Must Implement

The LOB application team must implement ONE callback class. The framework calls this class to decide whether reauthentication is required. Everything else — calling BTP, capturing signature, linking to Change Doc — is handled by the framework invisibly.

```
" Relevancy Check callback – LOB team implements this

CLASS zcl_po_reauth_check DEFINITION PUBLIC FINAL.

    PUBLIC SECTION.

        INTERFACES if_reauth_relevancy_check.

ENDCLASS.

CLASS zcl_po_reauth_check IMPLEMENTATION.

    METHOD if_reauth_relevancy_check~is_reauthentication_relevant.

        " Step 1: Identify which fields changed in this save

        " Step 2: Call framework utility to get configured regulated fields

        "             from S-table VC_SRA_SOT for PURCHASE_ORDER object

        " Step 3: Compare changed fields against regulated field list

        " Step 4: Return ABAP_TRUE if any regulated field changed

        "             Return ABAP_FALSE if nothing regulated changed

        "

        " WARNING: If this returns ABAP_FALSE due to a bug,

        "             reauthentication is SILENTLY SKIPPED

    ENDMETHOD.

ENDCLASS.
```

2.2 S-Table Configuration (SPRO → VC_SRA_SOT)

Register the PURCHASE_ORDER reauthentication object in S4 via SPRO. This content is delivered via scope item 84P transport — customer can fine-tune the regulated field list after activation.

SPRO → GxP Compliance → Reauthentication Settings → VC_SRA_SOT

Object:	PURCHASE_ORDER
Reauth Class:	ZCL_PO_REAUTH_CHECK ← LOB implements this

Auth Class: ZCL_PO_AUTH_CHECK ← LOB implements this

Node definitions:

Node: PO_HEADER_EXT

Table: /AT0/S4_PO_HEADER_EXT

Regulated Fields:

THERAPY_TYPE ← triggers reauthentication if changed

CLINICAL_STUDY ← triggers reauthentication if changed

PROTOCOL_REF ← triggers reauthentication if changed

Node: PO_ITEM_EXT

Table: /AT0/S4_PO_ITEM_EXT

Regulated Fields:

THERAPY_TYPE ← triggers reauthentication if changed

PATIENT_REF ← triggers reauthentication if changed

2.3 BTP Configuration — Reason Codes

Step	Activity	Who
1	Register PURCHASE_ORDER object in BTP eSignature SaaS	GxP Compliance Team
2	Define reason codes for PURCHASE_ORDER	GxP Compliance Team
3	Verify only PO reason codes appear in PO reauthentication dropdown	GxP Compliance Team
4	Suggested reason codes: Clinical Protocol Update, Regulatory Submission, Error Correction, Therapy Type Reassignment	LOB team proposes

Section 3 — Test Cases

Test cases are ordered by dependency. Test 1 must pass before Test 2. Tests 1–2 can be executed TODAY. Tests 3–5 require 2702 delivery.

Test 1 Verify Change Document for PO		■ Testable NOW
Test Steps	1. ME21N → create a new Purchase Order with any vendor and material 2. Save the PO — note the PO number (e.g. 4500000123) 3. ME22N → open the same PO → change a field (e.g. Delivery Date) 4. Save the change 5. ME23N → Environment menu → Changes OR SE16N → CDHDR → filter: OBJECTCLAS = EINKBELEG, OBJECTID = 4500000123 6. Check CDPOS for the same CHANGENR	
Verify	CDHDR: one row with OBJECTCLAS=EINKBELEG, USERNAME=your user, UDATE=today CDPOS: one row with FNAME=changed field, VALUE_OLD=old value, VALUE_NEW=new value	
Expected Result	Change Document created correctly — field change recorded with old and new values	

Test 2 Verify GxP Active Check (CHECK_ESIG_ACTIVE)		■ Testable NOW
Prerequisites	SE24 or ABAP Console access	
Test Steps	Run CHECK_ESIG_ACTIVE API from ABAP console or SE38: DATA lv_active TYPE abap_bool. " Call framework CHECK_ESIG_ACTIVE method " Log result to console Also verify in SPRO: SPRO → GxP Compliance → check scope item 84P activation status	
Verify	Result of CHECK_ESIG_ACTIVE call	
Expected Result	Returns ABAP_FALSE — EXPECTED in current state (Communication Arrangement not yet configured). Returns ABAP_TRUE only when: scope item 84P active AND BTP Communication Arrangement configured	

Test 3 Reauthentication Dialog Appears on Regulated Field Change		■ Requires 2702
Prerequisites	✓ Scope item 84P activated ✓ Communication Arrangement configured (S4 → BTP eSignature SaaS) ✓ ZCL_PO_REAUTH_CHECK implemented and registered in VC_SRA_SOT ✓ PURCHASE_ORDER registered in BTP with reason codes defined	
Test Steps	1. ME21N → create a new PO 2. Enter THERAPY_TYPE = CELL_GENE in the ATO Advanced Therapy Information tab 3. Click Save	
Verify	Reauthentication dialog appears before PO is committed	
Expected Result	Dialog shown with: - Password field (must re-enter — session token not sufficient) - Reason Code dropdown showing only PO-specific codes - Confirm and Cancel buttons	

Test 4 Signature Captured and Linked to Change Document		■ Requires 2702
Prerequisites	All prerequisites from Test 3	
Test Steps	1. Complete Test 3 — dialog appears 2. Enter correct SAP password 3. Select reason code: 'Clinical Protocol Update' 4. Click Confirm 5. PO saves successfully 6. Note the PO number and timestamp	
Verify	S4: SE16N → CDHDR → find CHANGENR for the PO CDPOS → FNAME=THERAPY_TYPE, VALUE_OLD="", VALUE_NEW='CELL_GENE' BTP: eSignature SaaS admin console → find Signature record: - Signature ID linked to CHANGENR - User = I308878 (your user) - Timestamp matches save time - Reason Code = Clinical Protocol Update - Credential hash present	
Expected Result	Complete GxP audit trail: CDHDR/CDPOS (what changed) + BTP Signature (who approved it intentionally)	

Test 5 Cancel Path — Save Must Be Fully Rolled Back		■ Requires 2702
Prerequisites	All prerequisites from Test 3	
Test Steps	1. ME21N → open a PO → change THERAPY_TYPE 2. Click Save → reauthentication dialog appears 3. Click CANCEL on the dialog	
Verify	S4: SE16N → CDHDR → confirm NO new entry for this PO at this timestamp S4: SE16N → /ATO/S4_PO_HEADER_EXT → confirm field value NOT changed ME21N: reopen the PO → confirm THERAPY_TYPE still has the original value	
Expected Result	PO save ABORTED — nothing committed to database. EKKO, EKPO, extension tables, CDHDR/CDPOS all unchanged. GxP integrity preserved — no partial save.	

Test 6 Non-Regulated Field Change — No Dialog		■ Requires 2702
Prerequisites	All prerequisites from Test 3	
Test Steps	1. ME21N → open a PO → change a NON-regulated field only (e.g. Purchase Order Text, Payment Terms) 2. Click Save	
Verify	No reauthentication dialog appears — PO saves immediately	
Expected Result	PO saved without dialog — Relevancy Check returned ABAP_FALSE. Change Document still created for the changed field. No signature record created in BTP — correct behaviour.	

Test 7 Unit Test — Relevancy Check Callback		■ Testable NOW (once class built)
Prerequisites	ZCL_PO_REAUTH_CHECK implemented in SE24	
Test Steps	Create ABAP unit test class for ZCL_PO_REAUTH_CHECK: Test Case A — Regulated field changed: Simulate: THERAPY_TYPE changed from "" to 'CELL_GENE' Expected: returns ABAP_TRUE Test Case B — Non-regulated field changed: Simulate: DELIVERY_DATE changed Expected: returns ABAP_FALSE Test Case C — No fields changed: Simulate: no field changes Expected: returns ABAP_FALSE Test Case D — Error in callback: Simulate: exception raised inside callback Expected: framework treats as FALSE — reauthentication skipped	
Verify	All 4 test cases pass in SE80 unit test runner	
Expected Result	All test cases pass — Relevancy Check behaves correctly for all scenarios	

Section 4 — Contacts & Current Status Summary

4.1 Key Contacts for Testing

Name	Role	Contact For
Swarnava Chatterjee	eSignature Integration Lead	BTP eSignature SaaS test system access, Communication Arrangement setup
Yadesh Gupta	Reauthentication Framework Architect	Framework APIs, S-table config, VC_SRA_SOT questions
Prabir Kumar Mallick	Reauthentication Framework Dev Lead	Technical implementation questions, BTP side config
Nils Hartmann	Purchase Order Product Owner	ME21N integration point, PO app questions
Satish Kumar	ATO/CGTO PM + Architecture	ATO extension fields, /ATO/S4_PO_HEADER_EXT questions

4.2 Test Execution Status Summary

Test	Description	Can Test Now?	Blocker
Test 1	Change Document for PO	■ YES	None — already verified 2608
Test 2	GxP Active Check (CHECK_ESIG_ACTIVE)	■ YES	None — will return FALSE (expected)
Test 7	Unit Test Relevancy Check Callback	■ YES*	*Once ZCL_PO_REAUTH_CHECK is built
Test 3	Reauthentication dialog in ME21N	■ NO	Needs 2702: ME21N integration + Comm Arrangement
Test 4	Signature captured + linked to Change Doc	■ NO	Needs 2702: BTP eSignature SaaS connected
Test 5	Cancel path — full rollback	■ NO	Needs 2702: ME21N integration
Test 6	Non-regulated field — no dialog	■ NO	Needs 2702: Relevancy Check integrated into ME21N

4.3 Delivery Roadmap

Release	Deliverable	Status
2608 (done)	Change Document for PO verified for all regulated fields	■ Complete
2702 (planned)	eSignature trigger in ME21N save flow	■ Planned
2702 (planned)	Communication Arrangement S4 ↔ BTP eSignature SaaS	■ Planned
2702 (planned)	Relevancy Check callback ZCL_PO_REAUTH_CHECK	■ Planned
2702 (planned)	S-table VC_SRA_SOT registration for PURCHASE_ORDER	■ Planned
2702 (planned)	BTP: Reason codes for PURCHASE_ORDER via 84P provisioning	■ Planned
2702 (planned)	ATO extension field registration in Change Document object	■ Planned — open issue
Post 2702	eSignature in modernised Manage PO App (BOF replacement)	■ Deferred